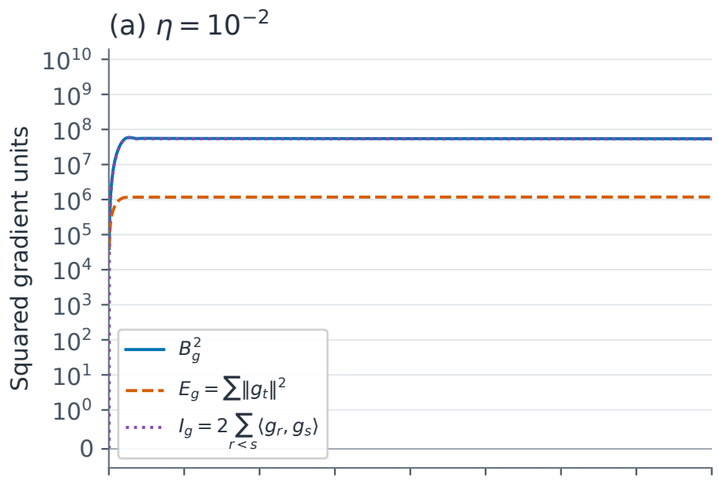
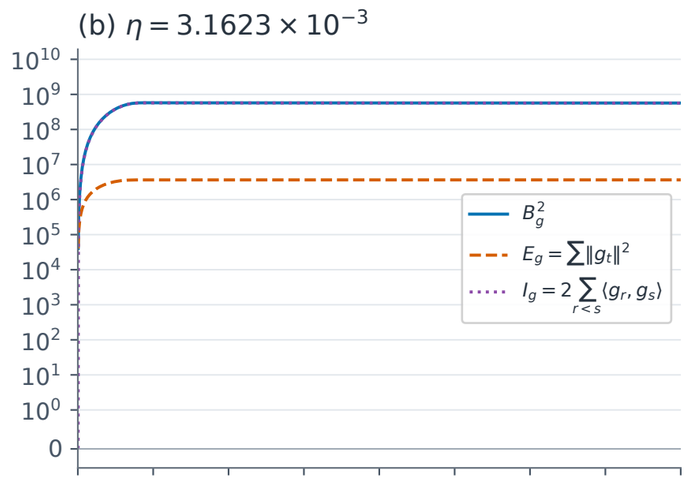
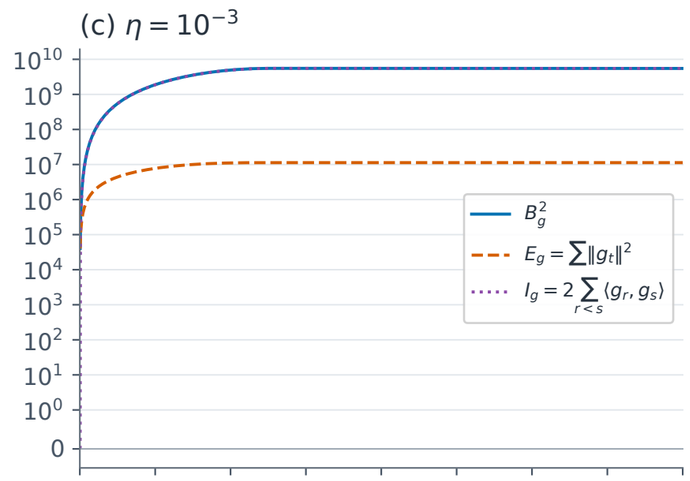
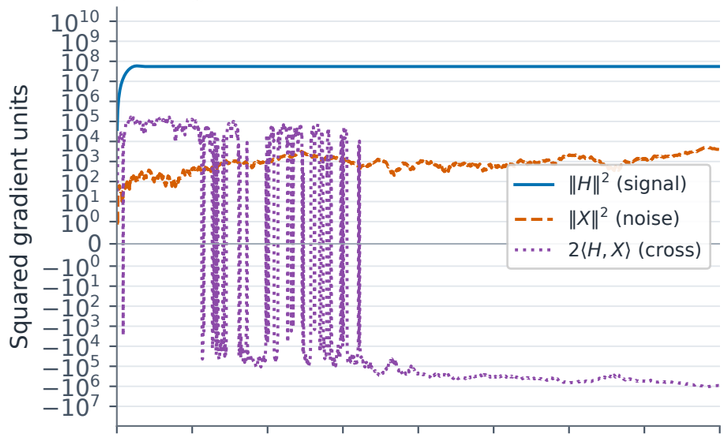
Rosenbrock noisy gradients · exact identities · σ = 1

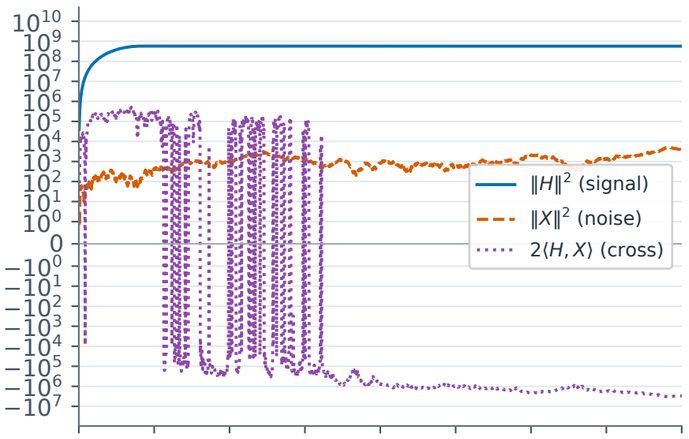
One paired noise seed (1234); every prefix shown; signed values use symmetric log Y (linear within ±1).

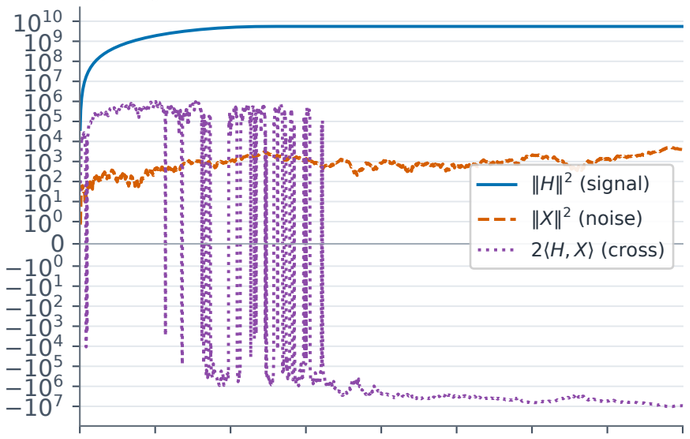


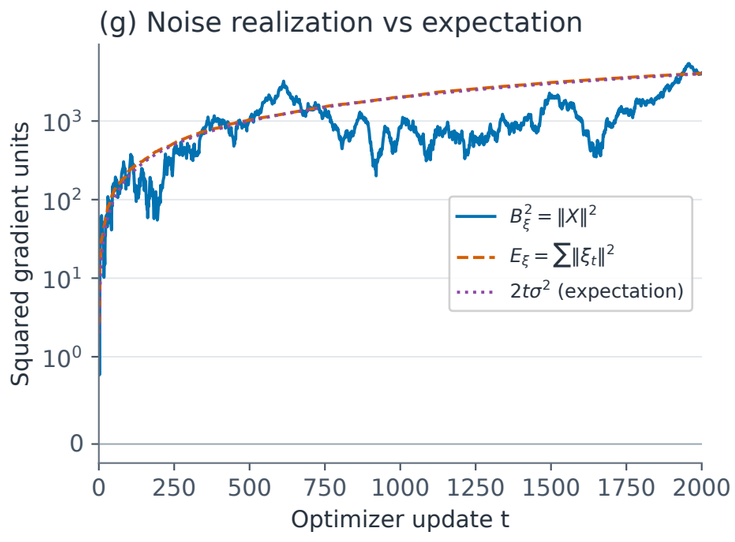


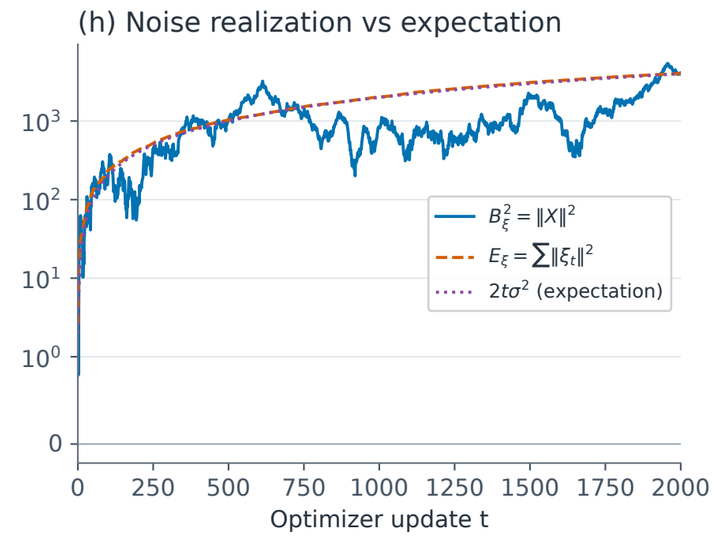


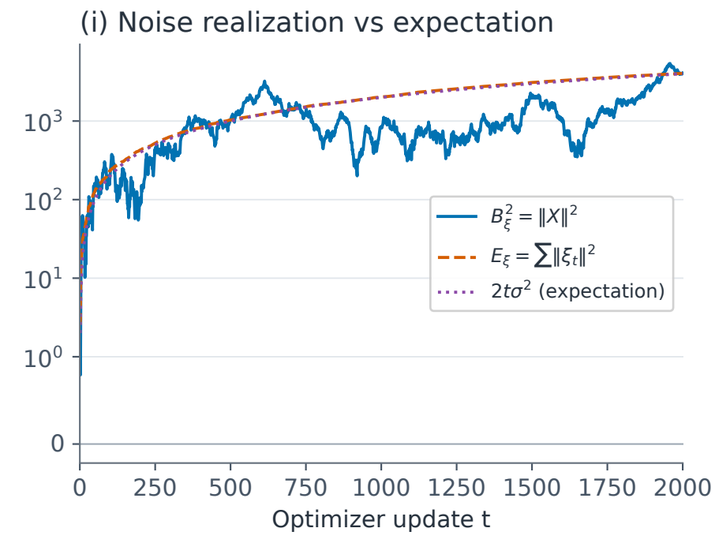












Exact: $B_g^2 = E_g + I_g = \|H\|^2 + \|X\|^2 + 2\langle H, X \rangle$, $H = \sum h_t$, $X = \sum \xi_t$, $h_t = \nabla f(\theta_{t-1})$.

The dotted ensemble expectation is not a pathwise identity. Cross terms remain on the adapted path; zero-mean noise does not set them to zero.